PARISHRAM 2.0 2026 (Class 12th)

Biology (Zoology)

DPP: 3

Human Reproduction

Q1 The process of formation of sperm from spermatogonia is known as:
(A) Spermiogenesis
(B) Spermatogenesis
(C) Oogenesis
(D) Gametogenesis

Q2 Acrosome is derived from:
(A) Mitochondria
(B) Golgi apparatus
(C) Nucleus
(D) Centriole

Q3 **Assertion (A):** Sertoli cells are also known as nurse cells.
Reason (R): They provide nourishment and support to developing spermatids.
(A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

Q4 Choose the correct option.
(A) Mitochondria - Neck
(B) Acrosome - Head
(C) Axial filament - Middle piece
(D) Nucleus - Tail

Q5 Choose the correct statement from the following.
(A) Spermatogenesis starts from the birth.
(B) Spermiation is conversion of spermatids into motile sperms.
(C) Spermatogonial cells divide mitotically and form primary spermatocyte.
(D) All of the above

Q6 If a spermatogonium (2n) undergoes 5 mitotic divisions before entering meiosis, how many primary spermatocytes will be produced?
(A) 32
(B) 64
(C) 128
(D) None of the above

Q7 Which part of the sperm contains enzymes that help in penetrating the ovum?
(A) Tail (B) Neck
(C) Acrosome (D) Middle piece

Q8 **Statement -1 :** GnRH a hypothalamic hormone stimulates secretion of LH and FSH.
Statement -2 : LH acts on Sertoli cells and stimulates secretion of androgens.
(A) Statement-1 is true statement-2 is false.
(B) Statement -1 is false statement-2 is true
(C) Both the statements are false.
(D) Both the statements are true

Q9 Which of the following is a diploid cell?
(A) Oogonium
(B) Secondary oocyte
(C) Ovum
(D) First polar body

Q10 Which of the following events distinguishes oogenesis from spermatogenesis?
(A) Meiotic divisions
(B) Unequal cytoplasmic division
(C) Presence of mitotic divisions

(D) Involvement of pituitary hormones

Answer Key

Q1	B
Q2	B
Q3	A
Q4	B
Q5	C
Q6	A
Q7	C
Q8	A
Q9	A
Q10	B

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 **Text Solution:**

The entire process of formation of spermatozoa (mature sperm cells) from spermatogonia (diploid stem cells) is called spermatogenesis.

Video Solution:

Q2 **Text Solution:**

The acrosome is a cap-like structure that covers the anterior part of the sperm head.It is formed from the Golgi apparatus during the process of spermiogenesis.

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Q3 **Text Solution:**

Both A and R are true, and R is the correct explanation of A.

Video Solution:

Q4 **Text Solution:**

Acrosome - Head

Video Solution:

Q5 **Text Solution:**

Spermatogenesis begins at puberty, not at birth. Spermiation is the release of mature spermatozoa from Sertoli cells into the lumen of seminiferous tubules.

Video Solution:

Q6 **Text Solution:**

Each mitotic division doubles the number of cells. So, if one spermatogonium (2n) undergoes 5 mitotic divisions, the number of resulting cells is:

$$2^5 = 32 \text{ cells}$$

Video Solution:

Q7 **Text Solution:**

The acrosome is a cap-like structure located at the anterior tip of the sperm head.It is derived from the Golgi apparatus and contains hydrolytic enzymes.

Video Solution:

Q8 **Text Solution:**

LH acts on Leydig cells, not Sertoli cells.

Video Solution:

Q9 **Text Solution:**

Secondary oocyte – Haploid (n), formed after meiosis I.Ovum –Haploid (n), formed only if fertilization occurs .First polar body – Also haploid (n), formed during meiosis I, non-functional.

Video Solution:

Q10 **Text Solution:**

Unequal cytoplasmic division - This is the key distinguishing feature of oogenesis not of spermatogenesis.

Video Solution:

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